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Spatial Potential Evaluation of Ecotourism Resources and Functional Zoning in the Qinghai Lake Basin.

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Abstract: Existing national park zoning studies exhibits significant gaps in integrating ecological sensitivity, tourism resource potential, and particularly the spatiotemporal dynamics of human activities. Against the backdrop of the Qinghai Lake National Park, which encompasses China's largest inland saltwater lake and serves as the core tourism region of Qinghai Province, this study develops a tri-dimensional evaluation framework of "place, people and things", where "place" - a multi-factors ecological sensitivity analysis, "people" - Mobile-data-based Active Population Estimation (MAPE) for spatiotemporal behavioral analysis and "things" - a comprehensive tourism resource potential assessment, and ultimately achieve functional zoning through multi-dimensional superposition. Key findings indicate that the overall ecological sensitivity of the basin is relatively high, with highly sensitive and extremely sensitive areas account for 46.24% of the total study area. Tourism potential has a highly uneven distribution, with limited-potential zones covering 42.90% of the basin, while top-level potential areas constitute merely 0.60%. Based on the integrated analysis, a refined five-zone functional system is proposed: strict protection zone, harmonious transition zone, innovative development zone, rational development zone, and stable maintenance zone, while the latter three account for less than 6%, revealing an extreme scarcity of “low ecological sensitivity, high tourism potential, and high tourist flow” areas. Each zone is coupled with differentiated control measures. This research provide quantitative support for the delineation of ecological protection redlines and sustainable tourism planning in Qinghai Lake National Park.

Keywords: Qinghai Lake Basin; Ecotourism Functional Zoning; Spatial Potential Evaluation; Spatiotemporal Behavior; MAPE;

1. Introduction

1.1 Research background

As the largest inland saltwater lake in China, Qinghai Lake is a typical representative of the lake wetland ecosystem on the Qinghai-Tibet Plateau. It also serves as the tourism center of Qinghai Province. Against the backdrop of the establishment of Qinghai Lake National Park, the ecotourism planning of the Qinghai Lake Basin is facing new opportunities and challenges: on the one hand, the establishment of Qinghai Lake National Park puts forward new concepts and requirements for the ecotourism planning in the basin, necessitating innovative approaches and pathways for traditional tourism models; on the other hand, the tourism industry contributes significantly to the GDP of the surrounding counties of Qinghai Lake, making it imperative to conduct scientific and comprehensive assessments to guide planning, regulation and tourism development.

The core concept of ecotourism is to take the natural beauty and regional characteristics and culture as the main tourism resources, to promote the protection of natural resources and cultural heritage, and to raise the awareness of tourists about the protection of natural resources and cultural resources, and ultimately realize the sustainable development of ecotourism. The term “ecotourism” was first introduced by the International Union for Conservation of Nature (IUCN) in 1983 and subsequently defined by the International Ecotourism Society in 1990. Correspondingly, ecotourism planning is an effective tool for the design, development and management of ecotourism sites, such as nature reserves or national parks, which are functionally zoned to achieve the dual objectives of ecotourism and nature conservation at the same time[1].

Current researches on ecotourism planning have drawn significant attention on ecotourism assessment indicators. Some studies constructed evaluation indicator systems to achieve effective assessment of either specific aspects or the overall framework of ecotourism[2]. In contrast to mass tourism that focuses on economic growth, ecotourism focused more on improving the quality of social, natural and human life[3-5]. Bhuiyan et al. discussed the perspectives including carrying capacity, economic benefits to local communities, and interaction between neighborhoods and tourist areas[6]. Specifically, community participation contributes to the sustainable development of the region by showcasing the unique local customs and cultural enrichment of ecotourism[7]. For instance, Samani et al. introduced the Limits of Acceptable Change (LAC) for sustainable ecotourism planning by identifying and evaluating the resources in the protected areas that are valuable to local stakeholders[8].

The functional zoning of tourism spatial functions and the delineation of development intensities have become crucial bases for coordinating the relationship between ecological environment and the protection and utilization of resources[9-11]. As an important part of the planning and management of national parks, zoning can relieve excessive focus on nature's benefits at the expense of human dimensions such as community development for indigenous people, tourists, etc.[12]. In terms of development strategies, there is a set of strict guidelines for ecotourism planning in national parks, including the ecotourism environment, resource assessment system, and measurement of environmental capacity for tourism. This is also a response to the public's demand for a high-quality natural resource and environment[13]. Xie et al. argued that better-educated individuals and those with income increase were more likely to have positive attitudes towards national park[14]. Shen et al. stressed that information sharing and timely response is beneficial for preventing conflict escalation between natural protection and livelihoods[15]. However, most existing ecotourism planning issues are simply proposing tourism development strategies from the perspective of problems in tourism development, and separately discuss the ecological security pattern and tourism resource sorting, which requires combining of the physical and human-behavior levels[10, 16-17]. In terms of methodology, ecotourism planning mainly relies on traditional GIS and expert judgment method. The applications of machine learning mainly on topics like tourism flow and path optimization [18], tourism emotion[19], tourists' preference[20], and landscape[21], with less in resource and function zone.

1.2 Research gaps

Existing scholarship on ecotourism planning and national park management has advanced significantly across multiple fronts, yet we argue that there are critical gaps in the integration of ecological indicators, tourism potential, and human behavioral dynamics.

1.2.1 National park zoning methods

Traditional approaches to national park zoning predominantly adopt a binary framework—dividing areas into “core protection zones” and “general control zones”[22-23]. This dichotomy prioritizes ecological conservation but often overlooks nuanced socio-ecological interactions, especially the varying intensities of human use and tourism pressure across landscapes [11]. While such zoning ensures strict protection in ecologically fragile regions, it lacks granularity for managing transitional or moderately sensitive areas where sustainable livelihoods and low-impact tourism could coexist with conservation goals. Moreover, the rigid separation between “protection” and “use” fails to accommodate dynamic feedback between ecosystems and human activities, limiting its applicability in complex area. Thus, it needs a multi-tiered, functionally differentiated zoning system that reflects both ecological thresholds and human dimensions.

1.2.2 Ecotourism resource evaluation

Ecotourism resource assessment has evolved from qualitative inventories toward quantitative, GIS-based evaluations[2, 24]. Recent studies emphasize indicators such as resource aggregation, dominance, and scale to gauge development potential[10]. However, most evaluations remain static—anchored to fixed geographic coordinates of attractions—and insufficiently account for spatiotemporal variability in actual tourist engagement. Consequently, current assessments risk overemphasizing physical resource density while underestimating functional tourism value shaped by real-world visitation patterns and community context.

1.2.3 Application of human behavior data

The incorporation of human behavior into spatial planning has long been constrained by data limitations. Traditional methods rely on surveys, census data, or expert judgment—sources that are either temporally sparse or lack fine-grained spatial resolution[16]. Although mobile big data have recently enabled more dynamic analyses of tourist flows[18-19], most existing models still treat human presence as a secondary overlay rather than a core dimension informing functional designation. Thus, leveraging real-time, high-resolution behavioral data could contribute to ecotourism zoning alongside ecological and resource criteria.

1.3 Study innovations

Address the gaps above, this study introduces three key innovations:

(1) Tri-dimensional integration. We construct a holistic evaluation framework that simultaneously incorporates ecological sensitivity (“place”), tourism resource potential (“thing”), and human spatiotemporal behavior (“people”). This triad enables a balanced consideration of environmental carrying capacity, resource endowment, and actual human use patterns—moving beyond siloed analyses toward systemic planning.

(2) Dynamic quantification of human activity via Mobile-data-based Active Population Estimation (MAPE). Unlike static demographic proxies, MAPE provides near-real-time insights into where and when tourists concentrate, thereby grounding zoning decisions in observed behavioral dynamics rather than assumed visitation.

(3) Refinement of binary zoning into a five-level functional system.

Departing from the conventional core/general dichotomy, we propose a more operable and context-sensitive five-zone typology—strict protection zones, harmonious transition zones, innovative development zones, rational development zones, and stable maintenance zones. This granular structure allows tailored management strategies that align ecological thresholds with tourism intensity and community needs, offering practical guidance for Qinghai Lake National Park implementation.

From the perspectives of people, places and things, this paper carries out a comprehensive study through ecological sensitivity analysis, tourism resource evaluation and spatiotemporal characteristic analysis using MAPE, and then propose functional zoning and control measures accordingly. By bridging these research gaps through methodological synthesis and empirical innovation, this study advances a more adaptive, evidence-based paradigm for ecotourism planning.

2. Research methods

This study proposed an integrated framework encompassing a three-step workflow of tourism area planning. Firstly, three categories of data were collected and integrated, namely “place” (e.g., topography, climate), “people” (e.g., population, tourist flows), and “things” (e.g., attractions), Secondly, ecological sensitivity analysis, MAPE clustering, and tourist development potential assessment were conducted, respectively, Ultimately, according to the results above, five functional zones were classified and corresponding measures were stated as well. The whole workflow diagram was shown in Figure 1.

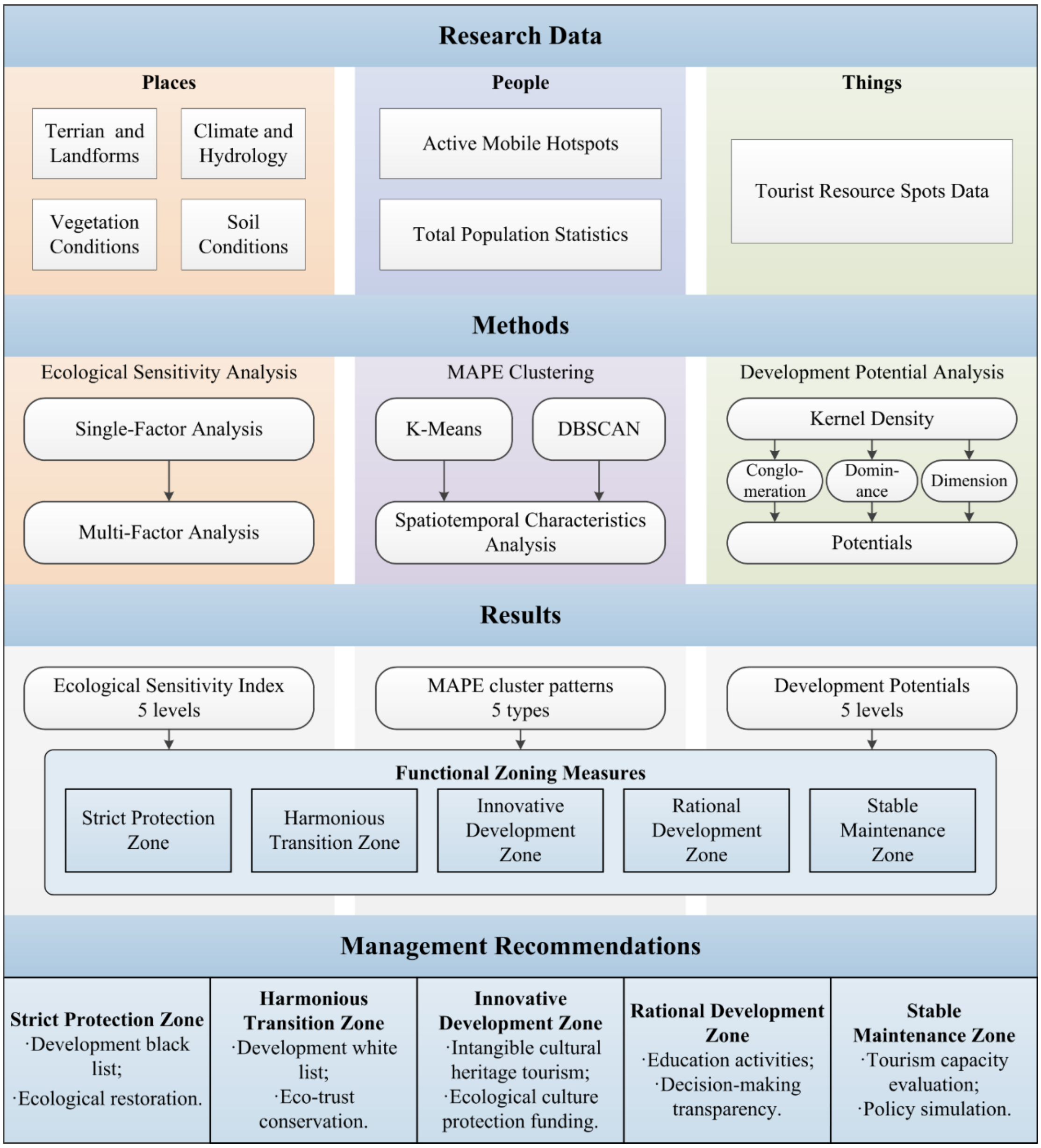
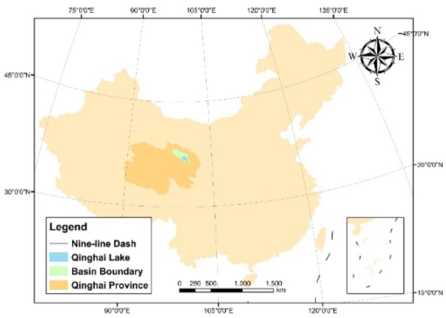


Figure 1. The workflow diagram of this study,

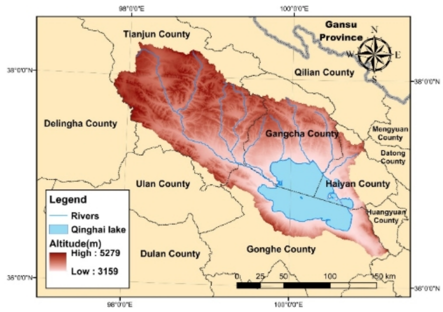
2.1 Study area and data sources

The Qinghai Lake basin (E97°48′55.70″–101°11′24.88″, N36°17′43.58″–38°19′16.20″) is located in the northeastern part of the Qinghai-Tibet Plateau, which is the largest inland saltwater lake in China. Figure 2(a) and 2(b) show the location and overview of the study area, respectively. The eastern part of the lake borders with Huangyuan County, bounded by the Riyue Mountain; the western part of the lake borders with the Qaidam basin, bounded by the Amuniniku Mountains; the northern part of the lake borders with the Datong River Basin, bounded by the Datong Mountains;

the southern part of the lake borders with the Chaka Salt Lake, bounded by the South Mountain of Qinghai. The total land area is 29661 km^2. The research area belongs to a plateau continental climate, with an average annual temperature of -4.6~4.0°C, average annual precipitation of 291~597mm, and average annual evaporation of 1300~2000mm[25].



(a) Location of the study area



(b) Overview of the study area

Figure 2. The location (a) and the overview (b) of the study area. The final figures were created by ArcGIS 10.7 (Esri, https://www.arcgis.com).

The research data encompass three categories: (1) Geographical data, including terrain and landform[26] (https://doi.org/10.11888/Terre.tpdc.303006), climate and hydrology[27] (https://doi.org/10.11888/Atmos.tpdc.302088), vegetation conditions[28-29] (https://doi.org/10.11888/Terre.tpdc.300328; https://doi.org/10.11888/Terre.tpdc.301126) and soil conditions[30] (https://data.tpdc.ac.cn/zh-hans/data/b5c18477-dcc2-4b8b-86b4-86fd3a8cbabb), provided by National Tibetan Plateau / Third Pole Environment Data Center.; (2) Tourism resource spots data, derived from a status quo survey with the support from the Qinghai Lake Scenic Area Protection and Utilization Management Bureau, as well as the Qinghai Provincial Culture and Tourism Department; and (3) Behavioral data, namely, the active mobile hotspot data provided by China Unicom System Integration Co., Ltd., Shaanxi Branch, and the total population data from the Qinghai Lake Scenic Area Protection and Utilization Management Bureau. The former is structured as hourly counts within grid cells across the Qinghai Lake basin during the peak tourism season.

2.2 Selection of ecological sensitivity evaluation indicators

The selection of ecological sensitivity evaluation indicators needs to consider the specific ecological issues and evaluation contents of the research area[31]. Combining with the actual situation of the Qinghai Lake basin such as high altitude, steep slope, plateau climate and Qinghai-Tibet Plateau plant subzone, a combined methodology of the AHP method, Delphi method, dominant factor method and entropy weighting method was operated on various text and image materials and remote sensing image data[32]. Ultimately, one target level of "comprehensive ecological sensitivity evaluation" and four criteria levels of terrain and landform, climate and hydrology, vegetation conditions, and soil conditions formed the ecological sensitivity evaluation system of the Qinghai Lake basin. (see Table S1 in Supplementary Materials)

The single-factor sensitivity index is calculated by Eq. 1[33]:

$$A_i = \sqrt[j]{a_{i1} \times a_{i2} \times \cdots \times a_{ij}} \qquad (1)$$

Where A_i - sensitivity index of the ith factor; $a_{i1} \sim a_{ij}$ - evaluation indicators corresponding to A_i; j - number of sub-level indicators.

In the analysis of multi-factor sensitivity, to highlight the impact of highly sensitive ecological issues on the comprehensive sensitivity of regional land resources, the principle of "barrel effect" is introduced. The sensitivity of an entire ecosystem often depends on its weakest factor, the more fragile the factor is, the greater its sensitivity. Thus, we used the extreme-value method, i.e., choose the most sensitive factor of each criterion level, to formulate the multi-factors sensitivity index[33] as Eq. 2:

$$I = \max(A_i) \qquad (2)$$

Where I - Multi-Factors Ecological Sensitivity Index; A_i - Sensitivity Index of the i th factor.

2.3 Machine-learning methods

2.3.1 Kernel density estimation

Kernel density estimation (KDE) is a method for determining the density of features in geographical space. Through kernel density estimation, the aggregation degree, aggregation areas, aggregation centers and the range of influence on the surroundings of features in geographical space can be expressed by graph visualization, making it a classic method for validating spatial distribution characteristics[34]. The kernel density estimation is conducted by Eq. 3:

$$f(s) = \sum_{i=1}^{n} \frac{1}{h^2} k\left(\frac{s-c_i}{h}\right) \qquad (3)$$

Where $f(s)$ - the kernel density estimation value; s - the density estimation point; n - the number of elements in the study area; h - the bandwidth, which is the searching radius; k - the weight function of the kernel; $(s - c_i)$ - the distance between kernel density centroid estimation points and POIs (points of interest). The larger the value of $f(s)$, the greater the clustering degree of the POI density in the geographical space, and the more aggregated the spatial distribution as well.

2.3.2 Density-based spatial clustering of applications with noise (DBSCAN) clustering analysis

DBSCAN clustering does not require pre-setting the number of clusters. It determines the boundaries of clusters based on the density of data points and automatically identifying the number of clusters[35]. The key advantage lies in its ability to identify clusters of any shape and effectively distinguish outliers (noise) that do not belong to any cluster. The algorithm mainly requires two parameters: the neighborhood radius (Eps) and the minimum number of points within the neighborhood radius to become a core object (MinPts). MinPts = k + 1, k = 2 × dimension - 1.

For the selection of Eps, an average nearest neighbor analysis and a multi-distance spatial clustering analysis are performed on tourist sites to solve the neighborhood radius from the geographical spatial perspective[36], and then convert into the corresponding longitude, with the conversion formula as Eq. 4:

$$\lambda = d \times \frac{360}{2\pi R\cos(\varphi)} \quad (4)$$

Where λ - longitude of the clustering feature; d - distance of the clustering feature; R - radius of the Earth; φ - latitude of the region.

2.3.3 K-Means clustering analysis

K-Means clustering is a widely used clustering technique and the most basic clustering algorithm. The core idea is to find the optimal partition of K clusters through iterations, minimizing the loss function under the clustering results[37]. Since the value of k is unknown, the elbow method is first used to determine the number of clusters. When performing cluster analysis, increasing the number of clusters will result in a more detailed grouping of samples, with each group of samples more closely clustered together. This process will gradually reduce the sum of squared errors (SSE), which represents the sum of the squared differences between the cluster centers and their inner points in cluster analysis. SSE is a key indicator used in the elbow method to evaluate the optimal number of clusters, calculated by Eq. 5:

$$SSE = \sum_{i=1}^{k} \sum_{x \in C_i} ||x - \mu_i||^2 \quad (5)$$

Where SSE - sum of squared errors; k - number of clusters; C_i - the set of samples in cluster i; x - a single sample point in cluster C_i; μ_i - the central point of cluster C_i (usually the mean or centroid of all points in the cluster); $||x - \mu_i||$ - the Euclidean distance from point x to centroid μ_i.

2.4 MAPE (Mobile-data-based Active Population Estimation) process

MAPE was carried out in two steps[38]. First, matching the active mobile hotspot data from China Unicom System Integration Co., Ltd., Shaanxi Branch, and total population data from the Qinghai Lake Scenic Area Protection and Utilization Management Bureau, to all grid cells of the study area. Through spatial intersection analysis, the expansion rate for active mobile hotspot counts towards total population can be calculated by Eq. 6:

$$K_\alpha = \frac{L_\alpha}{W_\alpha} \quad (6)$$

Where K_α - the expansion rate within cell α; L_α – active mobile hotspot count within cell α; W_α - total population count within cell α.

Second, estimate the total population within cell β during a given period τ. Where τ was set as one hour, corresponding to the scope of the active mobile hotspot counts. The estimated total population is calculated by Eq. 7:

$$Z_{\beta,\tau} = \sum \frac{I_{\beta,\tau}}{K_\alpha} \quad (7)$$

Where $Z_{\beta,\tau}$ - estimated total population within cell β and period τ; $I_{\beta,\tau}$ – active mobile hotspot count within cell β and period τ.

3. Multi-dimensional analysis of ecotourism in the Qinghai Lake Basin

3.1 Places: ecological sensitivity analysis of Qinghai Lake Basin

3.1.1 Single-factor sensitivity analysis

Using ArcGIS to visualize the single-factor ecological sensitivity index, the results were geometrically averaged and classified as 5 levels. As shown in Figure 3, under the terrain and landforms factor, the sensitivity decreases from the northwest mountains to the southeast lakes; the extremely sensitive areas under the climate and hydrological factors are concentrated in rivers, lakes and some parts of the northwest mountains; the sensitive areas under vegetation condition factor are mainly in the lake areas and western mountains; under the soil condition factor, there are more sensitive areas in the lakes and various mountain tops.

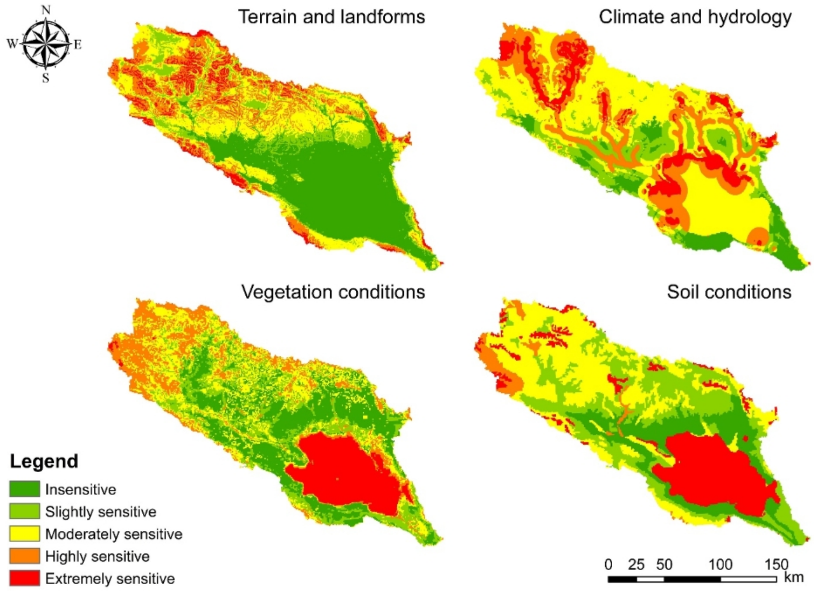


Figure 3. Single-factor Ecological Sensitivity Evaluation of Qinghai Lake Basin. There are 5 sensitivity levels for each factor, from 1 - Insensitive to 5 - Extremely sensitive. The final figures were created by ArcGIS 10.7 (Esri, https://www.arcgis.com).

3.1.2 Multi-factor Sensitivity Analysis

For each grid cell, the maximum level among the four single-factor sensitivity indices was taken as the multi-factor sensitivity level. The distribution map of ecological sensitivity levels in the Qinghai Lake basin and the corresponding statistical results is shown in Figure 4 and Table 1, respectively.

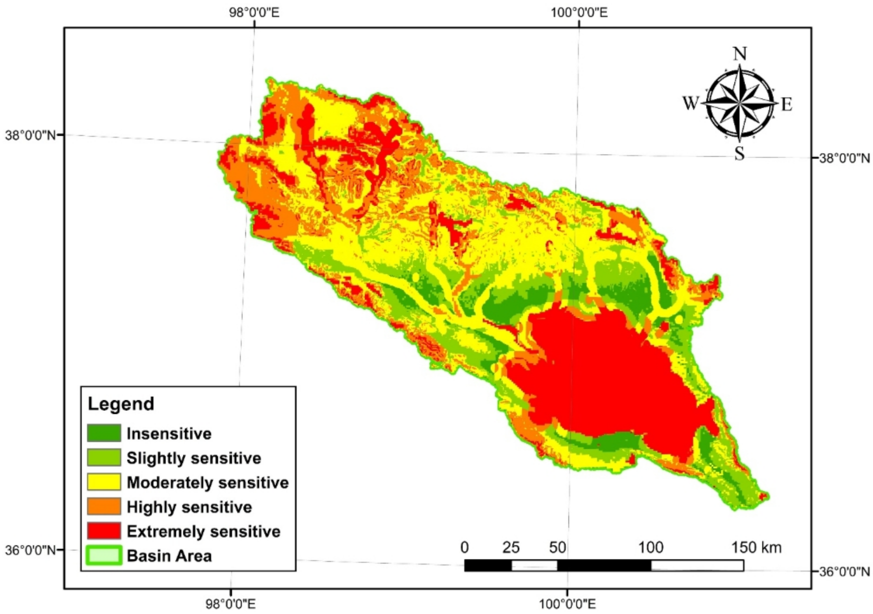


Figure 4. Multi-Factors Ecological Sensitivity Evaluation of Qinghai Lake Basin. The final figure was created by ArcGIS 10.7 (Esri, https://www.arcgis.com).

Table 1. Ecological Sensitivity of Qinghai Lake Basin.

Evaluation	Insensitive Area	Slightly Sensitive Area	Moderately Sensitive Area	Highly Sensitive Area	Extremely Sensitive Area
Area (km^2)	1972.46	4244.49	9722.88	5831.35	7883.89
Percentage (%)	6.65	14.31	32.78	19.66	26.58

The proportion of sensitive areas in the Qinghai Lake basin is relatively high, with highly and extremely sensitive areas covering nearly half of the area, which mainly distributed in the northern parts of Qinghai Lake and Tianjun County. Extremely sensitive areas are mainly concentrated in areas with low vegetation coverage, high elevation, steep slopes and the upper reaches of rivers and lakes in the basin. Moderately sensitive areas account for nearly one-third, connecting highly sensitive areas and slightly sensitive areas. In addition to the sensitivity caused by natural factors in the northern part of the basin, the characteristics of the county seats in several counties in the moderately sensitive areas also increase the impact of human activities in this region.

The ecological sensitivity of the study area is influenced by both natural and anthropogenic reasons. Since hydraulic erosion, wind erosion and soil erosion coexist in the Qinghai Lake Basin, areas with low vegetation cover, high elevation slopes, and the upper reaches of lakes and rivers are susceptible to hydraulic erosion and soil erosion, and thus they are extremely sensitive areas. Human activities are more obvious in highly sensitive areas and moderately sensitive areas, especially for the latter, whose distribution is basically within the radiation range of townships with relatively concentrated human activities. In contrast, slightly sensitive areas are mainly distributed in areas with higher vegetation cover, higher elevation and less human activities. Other than that, the trend of vegetation conditions, soil conditions and altitude slope are also positively correlated with the degree of ecological sensitivity.

3.2 Things: comprehensive evaluation of tourism resources in the Qinghai Lake Basin

3.2.1 Survey on the current situation of tourism resources

According to Classification, Investigation and Evaluation of Tourism Resources (GB/T 18972-2017) and Regulation of Resources Surveying and Evaluating in National Park (LY/T 3189-2020), a status quo survey of tourism resources in the Qinghai Lake basin was carried out. A total of 235 tourism resource spots in the Qinghai Lake basin had been identified and classified as two categories: natural and cultural tourism resources, as well as five levels (see Table S2 in Supplementary Materials).

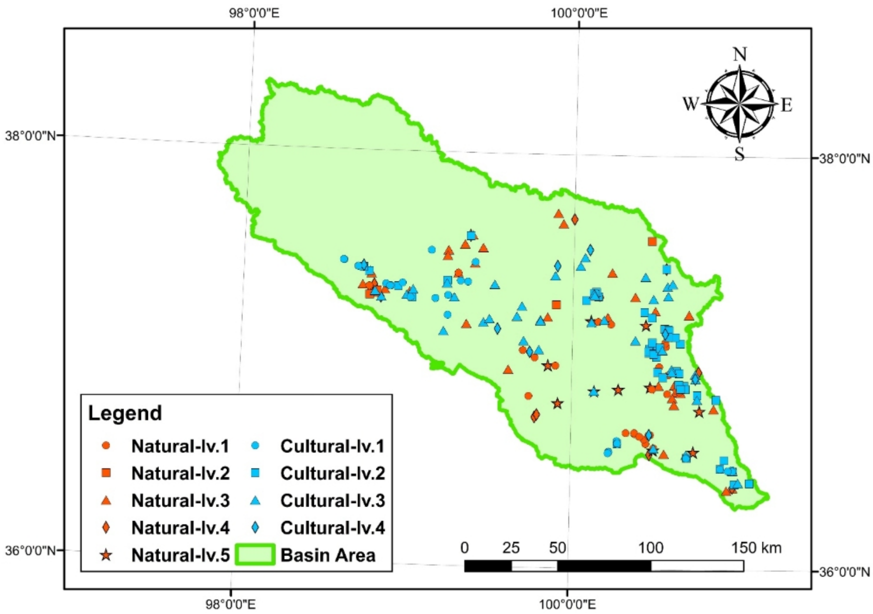


Figure 5. Distribution of tourism resource spots in the Qinghai Lake Basin. The final figure was created by ArcGIS 10.7 (Esri, https://www.arcgis.com).

As shown in Figure 5, the tourism resources in the Qinghai Lake basin are distributed unevenly in space, with a spatial distribution pattern of “aggregated around Qinghai Lake and dispersed along mountain water bodies”. The Qinghai Lake surrounding area serves as the core intensive area, with the Tianjun County, Tianjun Mountain and Zaxi County's Naishen Mountain in the secondary intensive areas. Moreover, tourism resources are distributed along major rivers like Buha River, Shaliu River, Hage River, and Quanji River, showing a clear hierarchical spatial distribution. What’s more, the northwestern of the Basin have no tourism resources due to the high ecological sensitivity and the difficulty of tourism development.

3.2.2 Evaluation of spatial aggregation of tourism resources area.

The kernel density analysis of all tourism resource units is shown in Figure 6, which suggests that the tourism resources within the basin are relatively rich and have formed obvious aggregations. The tourism resources in the basin have formed three major aggregation centers: Tianjun Mountain, Shaliu River and Gahai Lake, as well as two major aggregation sub-centers: Erlangjian Shelf and Daotang River.

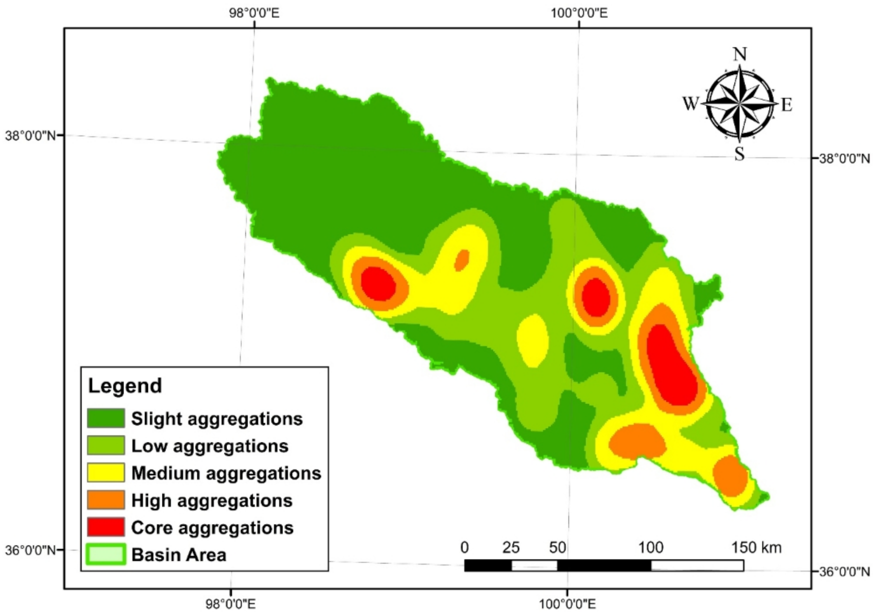


Figure 6. Distribution of tourism resources kernel density in the Qinghai Lake Basin. The final figure was created by ArcGIS 10.7 (Esri, https://www.arcgis.com).

To further identify the clustered spots and detect the discrete spots that do not have clustering features as well, all the spots were analyzed by DBSCAN clustering analysis, with the parameter Eps= 5400m, λ=0.054°. The tourism resource spots in the basin were divided into 19 clusters (Cluster 0-18) and discrete spots (Noise), as shown in Table 2.

Table 2. Representing tourism resource unit within each cluster.

Labels	Representative Tourism Resource Unit	Counts within the

		cluster
Cluster 0	Gesar King Ruins	7
Cluster 1	Tianjun Mountain, Tianjun Stone Forest	20
Cluster 2	Profiles of Caodigou-Xiahuancang Stratum	4
Cluster 3	Shaliu River Kayue Ruins, Lover's Cliff	18
Cluster 4	Xiaozhong Baifo Temple, Tongbao Mountain	4
Cluster 5	Gahai Ancient City Ruins	21
Cluster 6	Sand Island	10
Cluster 7	Przewalski's Gazelle Habitat	4
Cluster 8	Haiyan Bay	16
Cluster 9	Daotang River Mani Stone	6
Cluster 10	Chahan City	10
Cluster 11	Erhai Lake	6
Cluster 12	Dacang Ruins	5
Cluster 13	Hargan Mesa Ruins	4
Cluster 14	Zaxi County Naishen Mountain	6
Cluster 15	Buha River	5
Cluster 16	Seder Temple	4
Cluster 17	Fairy Bay (Xian’ nyu Wan) Wetland	6
Cluster 18	Erlangjian Shelf, South Mountain of Qinghai Lake	14
Noise	Shatuo Temple, Three Stone, Bird Island, Haixin Mountain	65

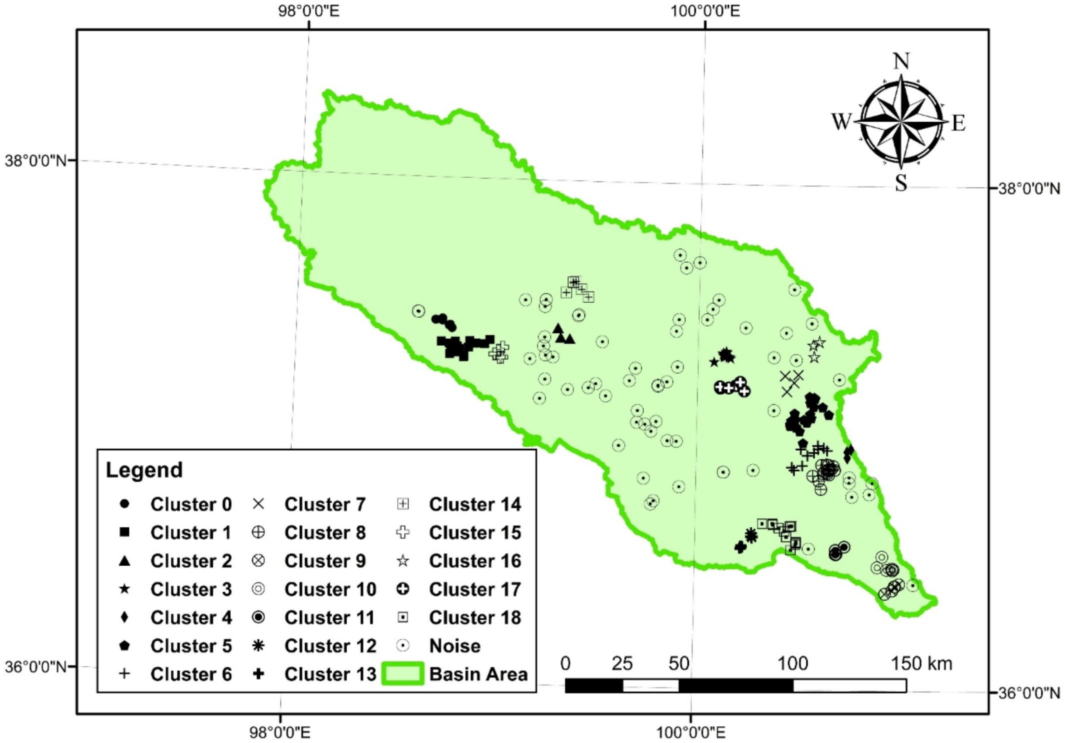


Figure 7. DBSCAN Clustering Results of Tourism Resources in Qinghai Lake Basin. The final figure was created by ArcGIS 10.7 (Esri, https://www.arcgis.com).

Figure 7 shows the distribution of the tourist spots within categories of 19 clusters and noises. Compared with kernel density analysis, clustering analysis can identify smaller tourist units, such as the aggregation center Gahai Lake, which is composed of 6 clusters including Tongbao mountain (Cluster 4), Sand Island (Cluster 6) and Przewalski's Gazelle (Procapra Przewalskii) habitat (Cluster 7) etc.; medium aggregation areas which formed by Shatuo Temple and Bird Island is composed of discrete tourism resource spots. The results indicate that Qinghai Lake's shoreline wetlands, grasslands, sand islands, and bird habitats form a high-value natural landscape foundation. Tourism infrastructure around Qinghai Lake has significantly enhanced accessibility, concentrating tourist activities around the lake area. As the Qinghai Lake basin is a convergence zone for multiple ethnic groups, their communication corridors primarily unfold along river valleys, resulting in cultural relics aggregated along water systems (such as the Shaliu River). Noises mostly scatter across moderately sensitive areas, where large-scale tourism development is restricted.

3.2.3 Evaluation of the development potential of tourism resources cluster

In the regional tourism resource system, tourism resources usually exist in the form of tourism resource clusters. Therefore, the development potential evaluation is carried out based on tourism resource clusters. Zhao et al. proposed that the development potential of tourism resource clusters is measured by the three main factors of aggregation, dominance of resource superiority index, and the size of regional tourism resource clusters, namely Conglomeration, Dominance and Dimensions[24]. This method is adopted in this paper as the evaluation index for assessing the development potential of tourism resource clusters in the Qinghai Lake basin.

(1) Conglomeration of regional tourism resource clusters

Extracting individual tourism resources units with clustering characteristics in the Qinghai Lake basin, kernel density analysis was conducted as the evaluation result of the clustering degree of tourism resources. Excluding scattered tourism resource units, eliminating their impact on regional kernel density could reduce unnecessary interference and improve the accuracy of the kernel density estimation. As is shown in Figure 8, The three major aggregation centers of Tianjun Mountain, Shaliu River, and Gahai Lake, alongside the two secondary aggregation subcenters of Erlangjian and Daotang River, exhibit a different shape and scale compared to Figure 7. Areas classified as highly aggregated or extremely aggregated have diminished, while non-aggregated zones have noticeably expanded. This findings indicate that tourism resources in the Qinghai Lake basin can be concentrated for development across three to five clusters to mitigate their impact.

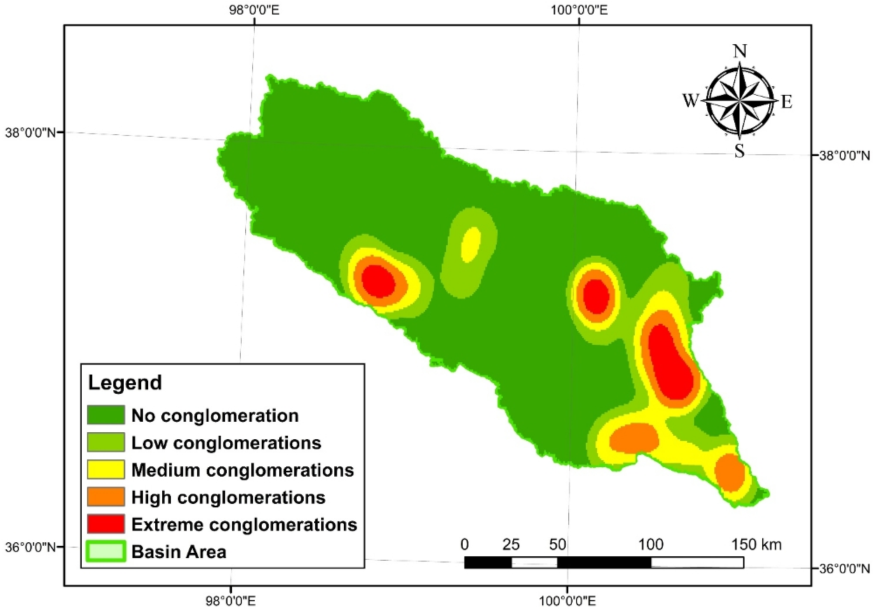


Figure 8. Conglomerations of Tourism Resources in Qinghai Lake Basin. The final figure was created by ArcGIS 10.7 (Esri, https://www.arcgis.com).

(2) Dominance of regional tourism resource clusters

To obtain the distribution of the dominance of regional tourism resource clusters, different buffer zones are established for tourism resource units obtained in the tourism resource survey according to five levels, and the weights of factors influencing each level are reasonably determined for overlay analysis. The weights of each A-level scenic spot are selected according to the research conclusions of Bai et al.[39], as shown in Table 3.

Table 3. Assessment results of development potential levels for tourism resources in the Qinghai Lake Basin.

Tourism resource spot level	1	2	3	4	5
Weight (%)	6.67	13.33	20.00	26.67	33.33

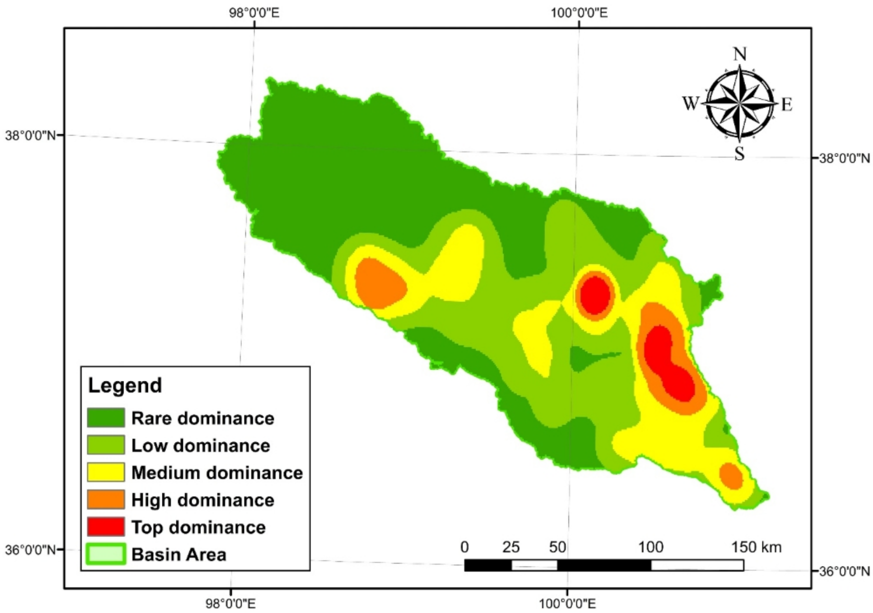


Figure 9. Dominance of tourism resources in the Qinghai Lake Basin. The final figure was created by ArcGIS 10.7 (Esri, https://www.arcgis.com).

The distribution of dominance is shown in Figure 9. Overall, the rare-dominant areas account for a relatively large proportion, while the top-dominant areas constitute a smaller proportion. The Shaliu River and Gahai formed two major top-dominant areas, while the Tianjun mountain and Daotang River areas are slightly inferior in quality, ultimately classified as high-dominant. Among them, the Shaliu River area has the top-dominant tourism resource entity of the Fairy Bay Wetland; the Gahai area has top-dominant tourism resource entities such as Shadao Island and the Przewalski's gazelle habitat, as well as multiple high-dominant tourism resource spots like Tongbao Mountain and Gahai Ancient City. High-dominant areas often have top-dominant tourism resource units or a relatively large number of high-dominant tourism resource units.

(3) Dimensions of regional tourism resource clusters

Measuring by the area of individual tourism resources, some data are difficult to obtain, and there is a natural difference in area between natural and cultural scenic spots. Therefore, the number of tourists is selected to approximate the dimensions. Using population as a weight field for analysis, kernel density analysis is used to show the spatial distribution of population and scenic spots.

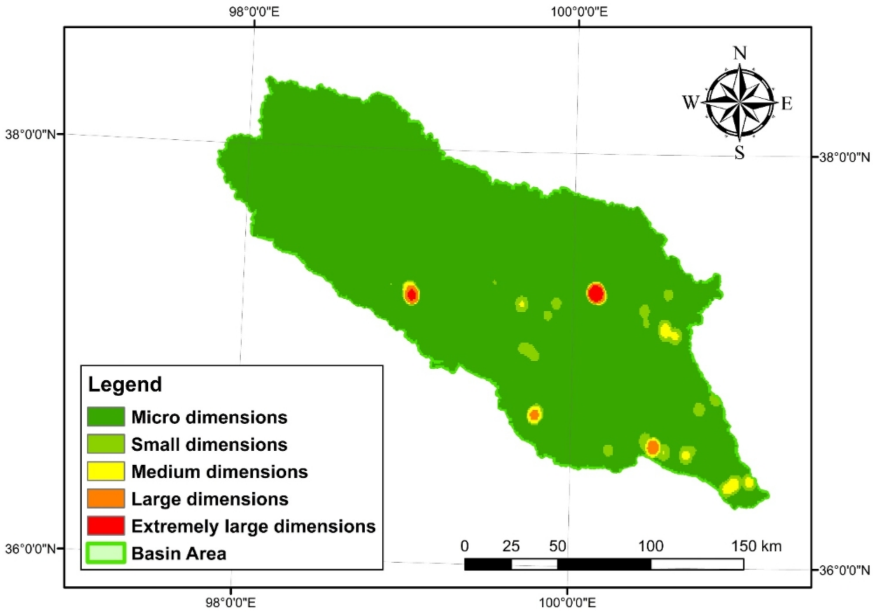


Figure 10. Dimensions of tourism resources in the Qinghai Lake Basin. The final figure was created by ArcGIS 10.7 (Esri, https://www.arcgis.com).

As is shown in Figure 10, Tianjun Mountain and Shaliu River area are extremely large dimensions areas; ErlangJian and Heima River area are large dimensions areas; Gahai, Daotang River and other areas are medium dimensions areas. Due to differences between natural resources and cultural resources, the dimensions vary significantly. Natural resources (such as Fairy Bay Wetland and Sand Island) typically attract large populations with strong visual impact and large space capacity. In contrast, cultural resources (such as ancient city ruins and temples) are often small and dispersed, thus difficult to gather large populations.

3.2.4 Comprehensive Development Potential of Regional Tourism Resource Cluster.

The geometric mean values of the conglomeration, dominance and dimensions of the tourism resources in Qinghai Lake basin are calculated to obtain the comprehensive development potential of the regional tourism resources (Figure 11). The top and high potential areas are mainly concentrated in the lakeside areas and the Tianjun Mountain region; lacking tourism resource units, the northern part of the basin shows nearly no development potential, presenting a large part of limited potential areas. Among all regions, the Shaliu River area and the Gahai Lake area demonstrate excellent performance in terms of conglomeration, dominance and dimensions, showing the top development potential. The Tianjun Mountain area, the Erlangjian area, and the Daotang River area have good conglomeration, high dominance and large dimensions, thus have high development potential. Other areas are classified as ordinary potential areas or basic potential areas due to their low conglomeration, low dominance or low dimensions (Table 4).

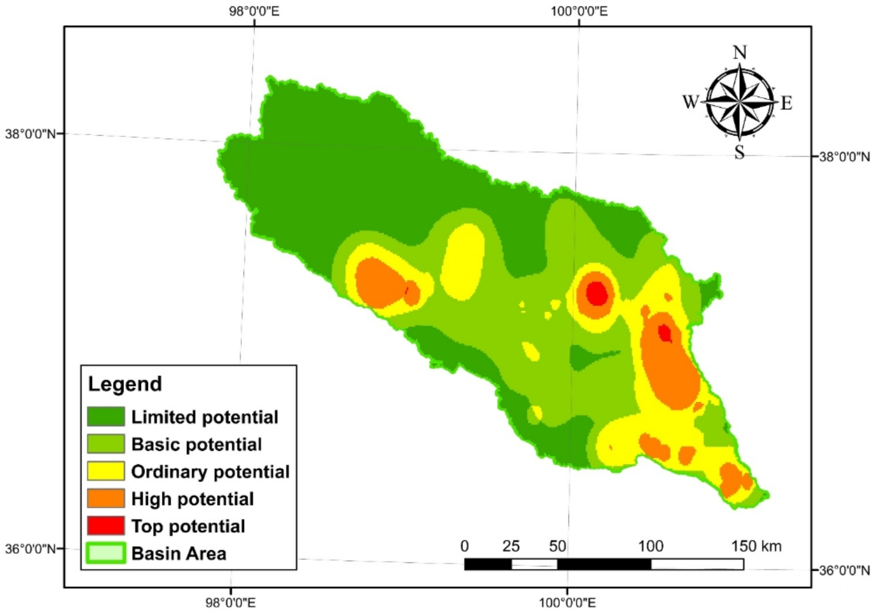


Figure 11. Distribution of the development potential of tourism resources in the Qinghai Lake basin. The final figure was created by ArcGIS 10.7 (Esri, https://www.arcgis.com).

Table 4. Assessment results of development potential levels for tourism resources in the Qinghai Lake Basin.

Evaluation	Limited potential area	Basic potential area	Ordinary potential area	High potential area	Top potential area
Area (km^2)	12723.93	9392.70	4801.78	2566.08	176.52
Percentage (%)	42.90	31.67	16.19	8.65	0.60

3.3 People: Spatiotemporal Characteristics Analysis of MAPE in Qinghai Lake Basin

MAPE showed the hourly averaged estimated total population within grid cells. To explore the spatiotemporal patterns of MAPE in the Qinghai Lake Basin, K-Means clustering analysis was conducted on the daily hourly pedestrian flow in all grid cells. Considering that there are 24 dimensions for each record, it was downscaled and then clustered in a low-dimensional space using t-distributed Stochastic Neighborhood Embedding(t-SNE), which is a popular nonlinear dimensionality reduction technique suitable for mapping high-dimensional data into 2D or 3D space[40]. Through elbow method and Silhouette method, the optimal number of clusters was determined as 7, with the Silhouette Coefficient as 0.81, revealing the cohesion of homo-clustered points and the separation of hetero-clustered points were both significant.

As shown in Figure 12, the distribution of MAPE of each cluster center was plotted for the 7 clusters, with the vertical axis representing pedestrian flow and the horizontal axis representing cluster class and time, revealing clear hierarchical patterns. The population size and diurnal variations are the most two primary factors distinguishing different clusters.

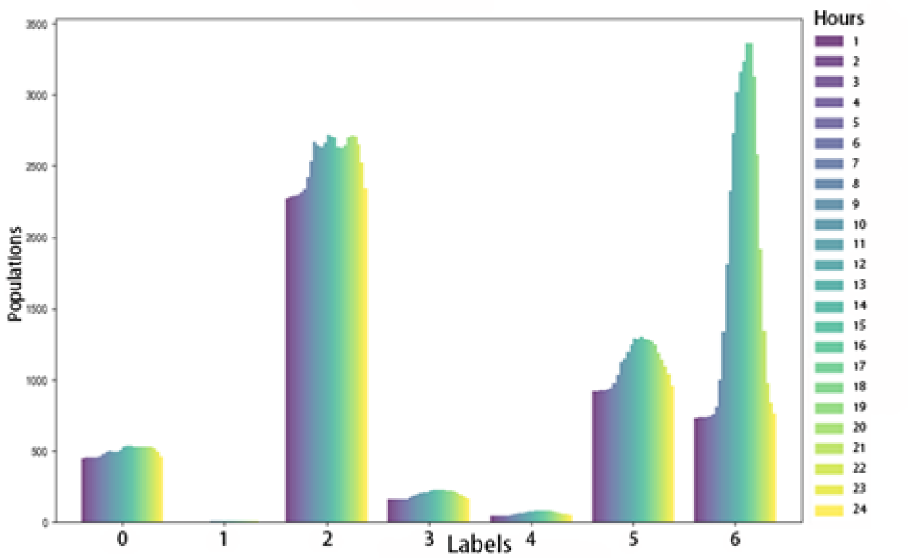


Figure 12. MAPE distribution of Cluster Centers.

Table 5. Labels and categories of Clusters.

Label	Category	Characteristics
0	Stable	Population size ≤ 500, similar population density during day and

	residential area	night.
1	Uninhabited area	Very low population size, often 0 or ≤ 9.
2	Daytime stable tourist area	Nighttime population size □ 2000, daytime population size □ 2500, significant day-night flow change.
3	Medium-sized residential area	Population size around 200, daytime peak population size around 250.
4	Low-density residential area	Population size lower than cluster 3, small-scale daytime growth.
5	Active tourist area	Nighttime population size around 1000, daytime increase in population flow, showing an inverted U-shaped pattern.
6	Peak tourist area	Nighttime population size lower than cluster 4, but rapid daytime growth, showing an inverted V-shaped growth, peak population flow □ 3000.

According to the clustering results, definitions of each cluster category were described in Table 5. Among them, medium-scale residential areas and low-density residential areas both have relatively small populations, with minimal impact on land resources and the environment, therefore merging into “small residential areas”. Active tourist areas and peak tourist areas both experience significant population growth during the day, demonstrating effective utilization of tourism resources and high popularity, therefore merging into “tourist hotspots”. Hence, we got 5 types of residential areas.

Figure 13 displays the spatial patterns of the 5 residential areas. The results indicate that most of the area is occupied by uninhabited area; small residential areas are distributed along the edges of Qinghai Lake, the coastline, Tianjun Mountain, and Daotang River; stable residential areas are sporadically scattered in small residential areas, which are relatively densely populated; tourist hotspots are very close to daytime stable tourist areas, and both are very close to government offices, which have good general infrastructure and public service facilities, making them preferred resting and stopping places for tourists.

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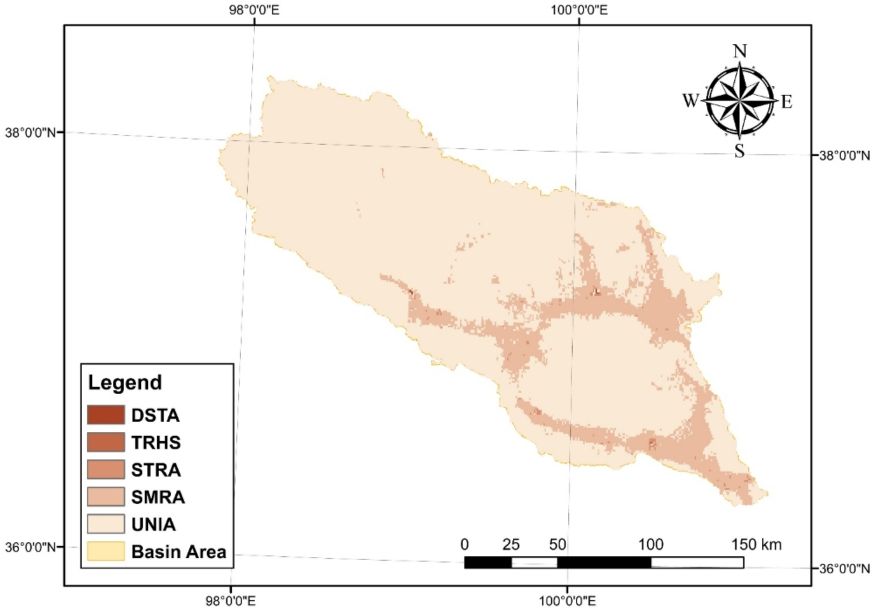

Figure 13. Spatial-Temporal Zoning Map of MAPE in Qinghai Lake Basin. Where DSTA is the Daytime Stable Tourist Area, TRHS is the Tourist Hotspot, STRA is the Stable Residential Area, SMRA is the Small Residential Area and UNIA is the Uninhabited Area. The final figure was created by ArcGIS 10.7 (Esri, https://www.arcgis.com).

4. Ecotourism Functional Zoning in the Qinghai Lake Basin

4.1 Construction of Ecotourism Zoning System in Qinghai Lake Basin.

Existing national park planning often divides the area into core protection zones and general controlled zones[22-23]. Among them, the core protection zone is the main body of the national park, with high ecological sensitivity, high conservation value, minimal human interference and the strictest protection measures implemented within the area; the general controlled zone is related to the residents' life and socio-economic development. It coordinates the public service functions such as scientific research, education, and ecotourism, with ecological protection as the top priority[11]. Based on the characteristics of national park zoning, this study subdivides the core protection zone into strictly protected areas and harmonious transition areas, while the general protection zone is divided into stable maintenance areas, reasonable development areas, and innovative development areas.

In the “Multi-planning Integration” functional zoning process of the

Qinghai Lake Basin, large gaps exist at level 2 (1-extremely sensitive to 5-insensitive) in ecological sensitivity and level 3 (1-limited potential to 5-top potential) in the development potential of tourism resource groups. Level 2 and 3 are then respectively taken as the boundary lines, as shown in Figure 14: (1) if the ecological sensitivity ≤ 2, then classifying as a strict protection zone; (2) if the ecological sensitivity > 2 and the tourism resources ≤ 3, with MAPE cluster tag be a residential area (STRA or SMRA) or uninhabited area, then classifying as a harmonious transition zone; (3) if the ecological sensitivity > 2 and the tourism resources ≤ 3, with MAPE cluster tag be a tourist area (DSTA or TRHS), then classifying as an innovative development zone; (4) if the ecological sensitivity > 2 and the tourism resources > 3, with MAPE cluster tag be a residential area or uninhabited area, then classifying as a rational development zone; (5) if the ecological sensitivity > 2 and the tourism resources > 3, with MAPE cluster tag be a tourist area, then classifying as a stable maintenance zone.

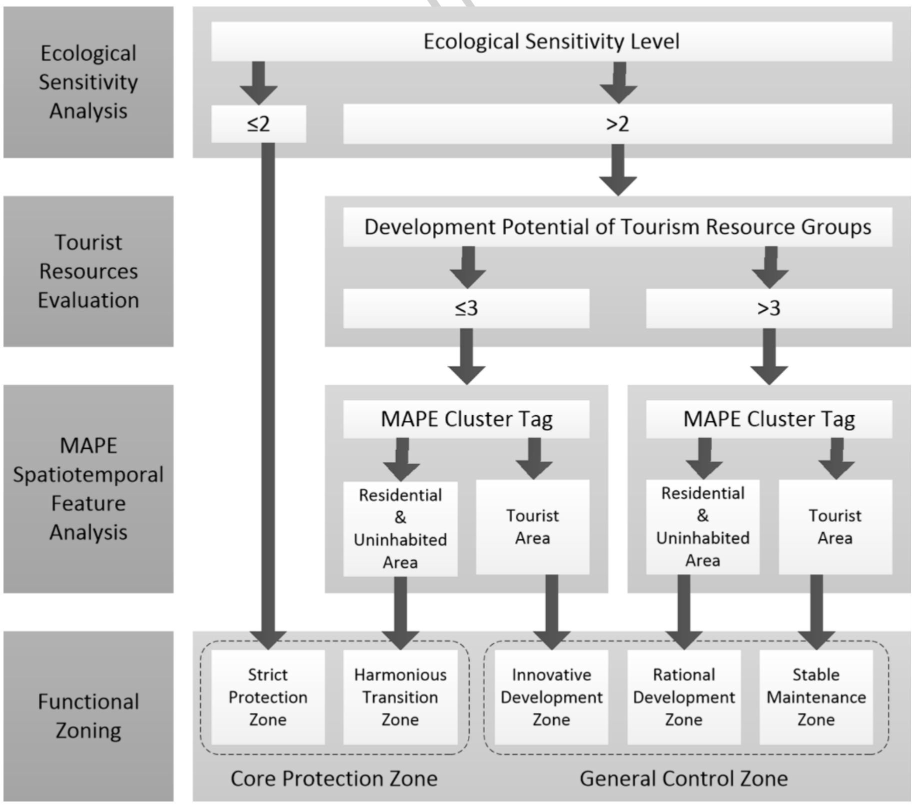

Figure 14. “Multi-planning Integration” Zoning process of the Qinghai Lake Basin.

Based on the above conditions, the ecological sensitivity, tourism development and MAPE distribution are overlaid in the multi-planning integration, as shown in Figure 15. The area and proportion of each zone are shown in Table 6. Generally, strict protection zones and harmonious transition zones have occupied most of the land; the proportion of reasonably developed areas mainly distributed in the Tianjun Mountain, Shaliu River, Gahai, Daotang River, and Erlangjian regions. The stable maintenance zones and innovative development zones are less distributed, with the former located in the government sites of Tianjun County and Gangcha County, and the latter in a small area in the east of the Sedar Line.

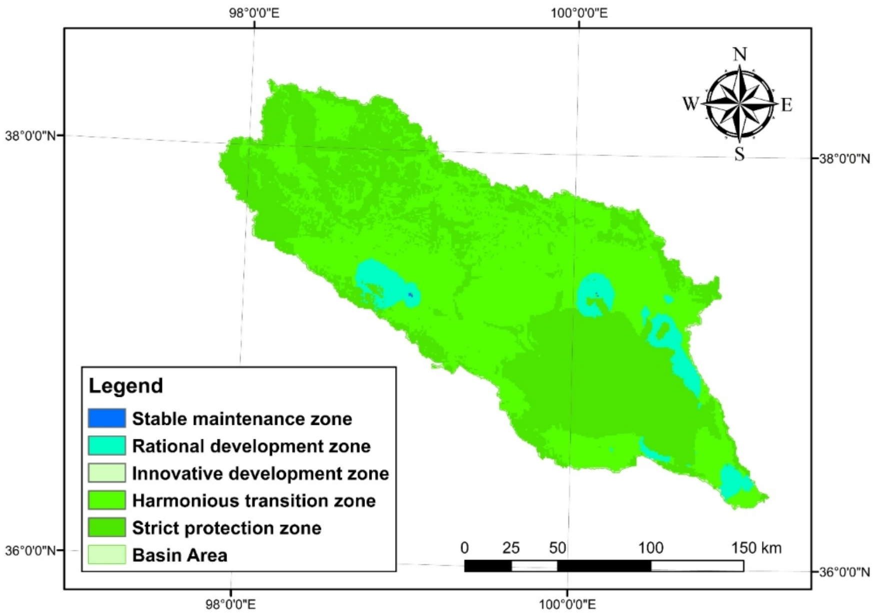

Figure 15. Ecotourism Functional Zoning of Qinghai Lake Basin. The final figure was created by ArcGIS 10.7 (Esri, https://www.arcgis.com).

Table 6. Statistical Table of Ecotourism Functional Zones in Qinghai Lake Basin.

Zones	Strict Protection Zone	Harmonious Transition Zone	Innovative Development Zone	Rational Development Zone	Stable Maintenance Zone
Area (km^2)	13868.14	14131.58	0.67	1655.23	5.38
Proportion (%)	46.76	47.64	2e-5	5.58	0.018

4.2 Refinement of the ecotourism zone types in the Qinghai Lake Basin

On the premise of ensuring the health of the Qinghai Lake basin ecosystem, the following suggestions are proposed for different regions in the Qinghai Lake basin to achieve a win-win situation for ecological protection and socio-economic development:

(1) Strict protection zone

The strict protection zone is highly ecologically sensitive, which requires the strictest protection measures to maintain the integrity and health of the ecosystem. Measures include using black lists for prohibiting large-scale development activities and human interference, carrying out ecological monitor system and regularly health examination for ecosystems, and promote ecological restoration projects, such as vegetation restoration, soil and water conservation, etc.

(2) Harmonious transition zone

The harmonious transition zone has a moderate ecological sensitivity, low population density, and exerts minimal pressure on the regional ecological environment. At the same time, it has low tourism potential and no further development needs. Measures include adopting limited white list for infrastructure construction and ecological forestry and pastoralism development, and piloting "eco-trust" model where a portion of tourism proceeds is directed to ecological conservation.

(3) Innovative development zone

The innovative development zone has low ecological sensitivity and limited tourism resources, but being identified as a tourist area means there is a significant demand for socio-economic activities. Due to the scarcity of areas meeting the criteria of low ecological sensitivity, low tourism potential and high tourist agglomeration, there is almost no innovative development zone. Still, it is necessary to discuss the control measures of these areas. The Qinghai Lake Basin boasts a rich intangible cultural heritage, including Thangka painting, Guozhuang dance, Tibetan opera, horse racing, etc., which can be utilized for cultural tourism project.

For example, setting up ecological culture protection funding to subsidize the living transmission of non-heritage skills, and transforming them into special tourism products.

(4) Rational development zone

The rational development zone has low ecological sensitivity and rich tourism resources, but has not yet formed a tourist area, providing unique opportunities for the sustainable development of the region. Measures include conducting ecological and environmental education activities towards tourists and residents, promoting community participation of residents and share the revenue through resource equity (e.g., B&B property rights, handicrafts), and applying digital hearing platform for controversial developments to enhance transparency in decision-making.

(5) Stable maintenance zone

The stable maintenance zone has low ecological sensitivity and abundant tourism resources, which has already formed a tourist area and a certain scale of tourism system but still requires a stable maintenance management strategy. They are also core nodes for tourism services. Measures include adopting limits of acceptable change and tourist behavior analysis to control the number of tourists, carrying tourism capacity evaluation regularly to prevent excessive tourism, and apply agent-based modeling toolkit for simulating decision-making under different policy scenarios.

5. Discussions

5.1 General discussions

From the perspectives of people, places and things, this paper aims to carry out a comprehensive ecotourism study through ecological sensitivity analysis, tourism resource evaluation and spatiotemporal characteristic analysis, and functional zoning and control measures accordingly. Against the backdrop of the Qinghai Lake National Park establishment program, this study directly contributes to the ecological baseline surveys and functional zoning work. Building upon the dichotomy of core protection zones and general controlled zones, the functional zoning of this paper has been further refined based on the degree of integration and emphasis placed on ecological protection and tourism development, though the lack of Innovative Development Zone and relatively small portion of Rational Development Zone and Stable Maintenance Zone challenge the validity of the zoning results. This may due to the overall high sensitivity of the Qinghai Lake Basin, especially the northwestern part of the basin, which emphasis the importance of the ecological protection over tourism development.

The resulting five-zone system - strict protection, harmonious transition, innovative development, rational development, and stable maintenance - is not merely a spatial overlay of three independent layers but a synthesis grounded in empirical evidence of how natural systems, resource endowments, and human dynamics interact on the ground. It is worth noting that tourism areas are not solely defined by the grid cells containing tourism resource spots, but rather by focal points where tourist attractions cluster, potentially offering more tourist services and better infrastructures. For instance, Residential areas also host numerous tourist attractions, though these are often of lower level, reflecting tourists' selective preferences. Uninhabited zones contain a few tourist attractions as well, which are similarly moderate level and frequently situated on mountain ranges, thus it is unsuitable for both residents and tourists.

Aside from the geographical indicators and tourism resource spots, there are still a large scale of intangible cultural heritage and other non-geographical resources. Though vital to place identity and visitor experience, they are difficult to integrate with our analysis due to their non-georeferenced nature[8]. For example, Thangka painting, Guozhuang dance, Tibetan opera and other traditional cultural activities have a relatively strong relationship with the residents, which may promote tourism potentials through the Stable Residential Area and the Small Residential Area defined in the MAPE analysis. Integrating spatiotemporal distribution of population and non-geographical tourism resource may help to promote residential participation in ecotourism and therefore improve socioeconomic benefits.

A pivotal contribution lies in the use of MAPE to quantify real-time tourist presence during peak seasons. The MAPE-derived activity maps reveal pronounced temporal peaks (e.g., midday concentrations at Fairy Bay and Erlangjian) and spatial hotspots that do not always align with official scenic area boundaries or high-density resource clusters. For instance, while Bird Island is a historically significant site, MAPE data show minimal visitor presence due to its closure for restoration, yet it remains classified as a high-value cultural resource in static inventories. Conversely, areas like Daotang River exhibit moderate resource density but attract sustained visitation due to accessibility and cultural resonance (e.g., Mani stone).

5.2 Research limitations

Despite its innovations, this study faces several methodological and contextual limitations, particularly when compared to recent advances in ecotourism and human mobility research.

First, MAPE data, while high-resolution in space and time, only capture China Unicom users, potentially underrepresent tourists from regions with lower Unicom penetration. This study requires a more robust population estimation frameworks, for instance, calibrating mobile data against census benchmarks[41].

Second, the static nature of the tourism resource inventory limits dynamic assessment. While we classified tourism resources regarding to GB/T 18972-2017 and LY/T 3189-2020 standards, the seasonal availability (e.g., bird migration windows at Qinghai Lake) or experiential qualities (e.g., scenic viewsheds, tranquility) are not considered. Other types of data could improve the seasonal availability of the ecotourism assessment[11].

Third, the five-zone typology, though more nuanced than binary zoning, still imposes categorical boundaries on continuous gradients of ecological and social variables. Conversely, fuzzy zoning or probabilistic land-use allocation using machine learning, allowing gradual transitions rather than sharp edges[17].

Finally, community perspectives were not directly incorporated. While MAPE captures tourist behavior, it does not reflect residents' livelihood dependencies or cultural values, which may risk overlooking sociocultural sensitivities[15].

5.3 Future directions

The future research can be deepened in the following aspects:

(1) Multi-source human mobility integration. Aside from mobile signaling data, other sources such as GPS traces from tourism apps and on-site counters can be complemented to produce a more representative activity panel.

(2) Dynamic resource–behavior coupling. Develop a time-aware evaluation system that adjusts tourism potential based on seasonal ecological conditions (e.g., breeding seasons for Przewalski's gazelle) and fluctuating visitor flows. This could enable monthly-shifting zoning measures.

(3) Community investigation for GIS-based planning. Conducting in-depth interviews and surveys on intangible cultural heritage within local communities, and integrate non-georeferenced knowledge and livelihood corridors into the zoning framework.

(4) Advanced machine-learning methods. Deep learning algorithms such as sparse CNN (Convolutional Neural Network), VSE-based (Variational Self-Encoder) zero sample learning and ALE-based (Attribute

Label Embedding) neural network could be applied to enhance the performance of the tri-dimensional evaluation framework of people, place and things.

6. Conclusions

This paper achieves a scientific ecotourism functional zoning of the Qinghai Lake basin through integrated ecological sensitivity analysis, comprehensive evaluation of tourism resources, and spatiotemporal characteristic analysis of MAPE, thereby promoting the ecological environment protection and sustainable development of the tourism industry in the Qinghai Lake basin.

(1) In terms of ecological sensitivity analysis, nearly half of the Qinghai Lake basin is a highly sensitive and extremely sensitive area, mainly in the northern part of Qinghai Lake and Tianjun County, which is closely related to natural factors such as high altitude, steep slopes, low vegetation cover and the headwaters of rivers and lakes. An ecological protection red line needs to be delineated to safeguard the integrity of the ecosystem.

(2) Through comprehensive evaluation of tourism resources, the core clusters of tourism resources are distributed around Qinghai Lake, and the sub-clusters extend along major rivers such as Buha River and Shaliu River. Through kernel density estimation and DBSCAN cluster analysis, 19 tourism resource clusters (such as Shaliu River Fairy Bay Wetland, Gahai Ancient City, etc.) were identified, among which Shaliu River and Gahai areas have the highest development potential.

(3) A spatiotemporal analysis of MAPE using K-Means has been applied. The results showed a significant spatial heterogeneity. Using K-Means clustering combined with t-SNE downscaling technique, the region was divided into five types of typical spatial units, namely, uninhabited area, small residential area, stable residential area, the tourist hotspot and the daytime stable tourist area.

(4) By integrating ecological sensitivity analysis, comprehensive

evaluation of tourism resources, and MAPE spatiotemporal characteristic analysis, the Qinghai Lake Basin is divided into five function zones, namely strict protection zone, harmonious transition zone, innovative development zone, rational development zone and stable maintenance zone. Differentiated management strategies are proposed for each zone.

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Data availability

The datasets generated and/or analysed are available from the corresponding author on reasonable request.

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